
AVR Interfacing

Interrupt

Agenda

- Interrupts Definition
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 - AVR External Interrupts
 - AVR External Interrupts Programming
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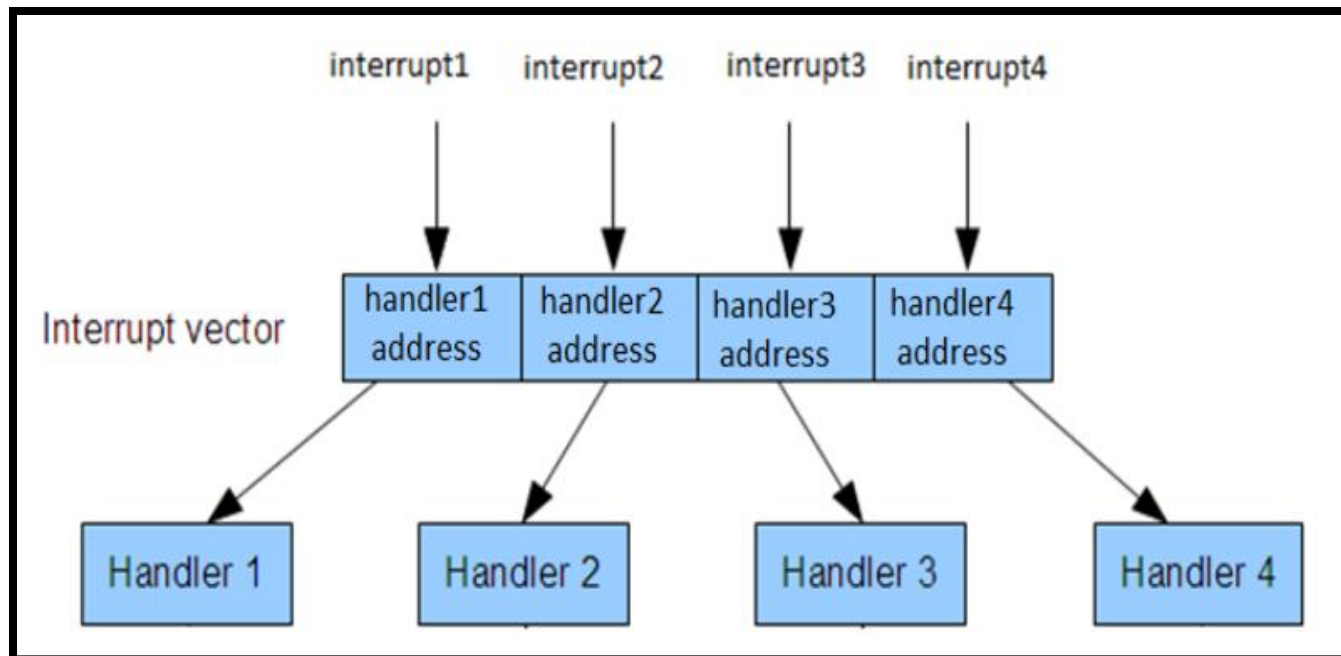
Interrupts Definition

- An interrupt is a signal to the processor emitted by hardware or software indicating an event that needs immediate attention.
 - When an interrupt event occurs, the microcontroller pause its current task and attend to the interrupt by executing an **Interrupt Service Routine (ISR)** at the end of the ISR the microcontroller returns to the task it had pause and continue its normal operations.
 - **Interrupt Service Routine (ISR) or Interrupt Handler**
Piece of code that should be execute when an interrupt is triggered.
 - Each interrupt has its own ISR. Its address in ROM is save in interrupt vector table.
 - It should be deterministic and short as possible so not to pause the CPU much of time.
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Interrupts Definition

- **Interrupt Vector Table (IVT)**

- Constant table in Flash EEPROM Program memory.
- Each interrupt has specific address in the interrupt vector table for its ISR.
- **Handler = ISR = Code**



AVR Interrupts

- **Interrupt Vector Table (IVT) of Atmega328P**

- The lower the address the higher is the priority level.
- All interrupts are assigned individual enable bits to which must be written a logic one together with the Global Interrupt Enable bit

ATmega328P		
Vector Number	Interrupt definition	Vector name
2	External Interrupt Request 0	INT0_vect
3	External Interrupt Request 1	INT1_vect
4	Pin Change Interrupt Request 0	PCINT0_vect
5	Pin Change Interrupt Request 1	PCINT1_vect
6	Pin Change Interrupt Request 2	PCINT2_vect
7	Watchdog Time-out Interrupt	WDT_vect
8	Timer/Counter2 Compare Match A	TIMER2_COMPA_vect
9	Timer/Counter2 Compare Match B	TIMER2_COMPB_vect
10	Timer/Counter2 Overflow	TIMER2_OVF_vect
11	Timer/Counter1 Capture Event	TIMER1_CAPT_vect
12	Timer/Counter1 Compare Match A	TIMER1_COMPA_vect
13	Timer/Counter1 Compare Match B	TIMER1_COMPB_vect
14	Timer/Counter1 Overflow	TIMER1_OVF_vect
15	Timer/Counter0 Compare Match A	TIMER0_COMPA_vect
16	Timer/Counter0 Compare Match B	TIMER0_COMPB_vect
17	Timer/Counter0 Overflow	TIMER0_OVF_vect
18	SPI Serial Transfer Complete	SPI_STC_vect
19	USART Rx Complete	USART_RX_vect
20	USART Data Register Empty	USART_UDRE_vect
21	USART Tx Complete	USART_TX_vect
22	ADC Conversion Complete	ADC_vect
23	EEPROM Ready	EE_READY_vect
24	Analog Comparator	ANALOG_COMP_vect
25	Two-wire Serial Interface	TWI_vect
26	Store Program Memory Read	SPM_READY_vect

AVR Interrupts

- **What happens when an interrupt occurs in AVR Microcontrollers?**
 - The microcontroller completes the execution of the current instruction.
 - Clears the Global interrupt enable bit.
 - Stores the address of the next instruction that should have been executed (the content of the **PC**) and all the CPU registers are pushed onto the stack.
 - The interrupt vector of the triggered interrupt (ISR start address of this interrupt) is then loaded in the **PC(program counter)** from the **interrupt vector table**.
 - Microcontroller starts execution from that point up until reaches the end of the **ISR**.
 - The address that was stored on the stack in **step 1** is reloaded in the **PC** register.
 - The Global interrupt enable is re-enabled.
 - The micro-controller then continue executing the program.
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AVR Interrupts

- **Status Register**

Contains the status of the flags such as Overflow flag, Negative flag, Zero flag, Carry flag, Half-carry flag, **Global Interrupt mask (I) bit**.

Bit	7	6	5	4	3	2	1	0	
	I	T	H	S	V	N	Z	C	SREG
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- **How to set the I-bit?**

$\text{SREG} = \text{SREG} | (1 \ll 7);$

- **How to clear the I-bit?**

$\text{SREG} = \text{SREG} \& (\sim(1 \ll 7));$

- On Atmega328P microcontroller there are two (2) external interrupts:
 - **External Interrupt 0 (INT0)** : Triggered from pin 2 (PD2).
 - **External Interrupt 1 (INT1)** : Triggered from pin 3 (PD3).

AVR External Interrupts Programming

- **External Interrupt Mask Register**

The ATmega 328P supports two external interrupts which are individually enabled by setting bits INT1 and INT0 in the External Interrupt Mask Register

Bit	7	6	5	4	3	2	1	0
0x1D (0x3D)	-	-	-	-	-	-	INT1	INT0
Read/Write	R	R	R	R	R	R	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

- **EICRA – External Interrupt Control Register A**

The external interrupt control register A contains control bits for interrupt sense control

Bit	7	6	5	4	3	2	1	0
(0x69)	-	-	-	-	ISC11	ISC10	ISC01	ISC00
Read/Write	R	R	R	R	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

EICRA

AVR External Interrupts Programming

- **EICRA – External Interrupt Control Register**
 - INT0 control

ISC01	ISC00	Description
0	0	The low level of INT0 generates an interrupt request.
0	1	Any logical change on INT0 generates an interrupt request.
1	0	The falling edge of INT0 generates an interrupt request.
1	1	The rising edge of INT0 generates an interrupt request.

- INT1 control

ISC11	ISC10	Description
0	0	The low level of INT1 generates an interrupt request.
0	1	Any logical change on INT1 generates an interrupt request.
1	0	The falling edge of INT1 generates an interrupt request.
1	1	The rising edge of INT1 generates an interrupt request.

